Identification of a bistable normal form for regime transitions in a driven dissipative transport system

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Abstract

Spatially extended dissipative systems under strong forcing can switch abruptly between distinct macroscopic transport regimes. We propose a hybrid reduced-order model in which a one-dimensional diffusion equation with a state-dependent transport coefficient is coupled to a scalar latent variable governed by the cusp normal form $\tau\,\dot{\zeta} = a(\boldsymbol{w}) + b\,\zeta - \zeta^3$. The two stable branches of the cusp define the two transport regimes; the transition and its hysteresis are saddle-node bifurcations of the latent dynamics, and the distance of the exogenous drive from the fold provides a quantitative, closed-form margin to transition. All model components — the transport coefficient, a residual source term, the drive map, and an observation map from the latent state to a measured emission proxy — are identified end-to-end from experimental time series by differentiating through an implicit–explicit integrator. Unlike proxy-following approaches, the latent dynamics are driven exclusively by exogenous actuation, so the inferred regime state is falsifiable against held-out observables. The identified model assigns a calibrated regime probability at every instant, locates transition times without supervision beyond weak segment labels, and yields bifurcation diagrams that summarize when a transition can be forced or undone.

1 Introduction

Abrupt transitions between metastable states are a recurring motif in nonequilibrium systems: bistability, hysteresis, and critical slowing down near folds are generic features predicted by normal-form theory (Kuznetsov, 2004; Strogatz, 2018). In the experimental system considered here, a driven dissipative transport process switches between a low-transport-barrier regime (*state* L) and a high-transport-barrier regime (*state* H) as the external drive increases, with the reverse switch occurring at a lower drive — the signature of hysteresis. A quantitative, model-based answer to the question *"which regime is the system in, and how far is it from switching?"* is a prerequisite for control.

Our contributions are: (i) a hybrid mechanistic–statistical model in which the regime is a latent coordinate of a cusp normal form, giving the L/H classification a precise mathematical meaning (basin membership of a bistable vector field); (ii) end-to-end identification through a differentiable implicit–explicit solver; and (iii) closed-form bifurcation diagnostics — fold margins $\mu_{\mathrm{LH}}(t)$ and $\mu_{\mathrm{HL}}(t)$ — that quantify proximity to forced and spontaneous transitions.

2 Model

2.1 Transport backbone

Let $u(\rho, t)$ denote the transported scalar field on the normalized domain $\rho \in [0, 1]$ with Neumann condition at $\rho = 0$ and a measured Dirichlet trace at $\rho = 1$. The field obeys a conservative diffusion law with a learned residual source s_θ,

$$\partial_t u = \frac{1}{V'(\rho)}\,\partial_\rho\big(V'(\rho)\,\chi(\rho,\zeta)\,\partial_\rho u\big) + s_\theta(\rho,u,n,\boldsymbol{w},\zeta)\,, \tag{1}$$

where V' is a fixed geometric weight and n an auxiliary measured field. The transport coefficient is a monotone boundary-layer profile whose edge value is modulated by the latent regime coordinate ζ,

$$\chi(\rho,\zeta) = \chi_0 + \sigma\big(\tfrac{\rho-\rho_c}{w}\big)\,[\chi_1 - \Delta\chi\,\beta(\zeta) - \chi_0]\,, \qquad \beta(\zeta) = \sigma\Big(\tfrac{3\zeta}{\sqrt{b}}\Big)\,, \tag{2}$$

so that the H regime ($\zeta > 0$) suppresses transport in the boundary layer and steepens the stationary profile there.

2.2 Latent regime dynamics: the cusp normal form

The regime coordinate evolves according to

$$\tau\,\dot{\zeta} = a(\boldsymbol{w}) + b\,\zeta - \zeta^3, \qquad b > 0, \tag{3}$$

with drive $a(\boldsymbol{w}) = \boldsymbol{\kappa}^\top \boldsymbol{w} + \kappa_0$ an affine map of the exogenous inputs only (injected power, current, mean density). For $|a| < a_\mathrm{f} = 2(b/3)^{3/2}$ the vector field (3) is bistable: stable equilibria $\zeta^*_\mathrm{L} < 0 < \zeta^*_\mathrm{H}$ are separated by a saddle ζ_s. Sweeping a through $+a_\mathrm{f}$ annihilates the L branch in a saddle-node bifurcation (forced L→H transition); sweeping back through $-a_\mathrm{f}$ annihilates the H branch (H→L back-transition). Hysteresis is therefore intrinsic rather than imposed. We define the *fold margins*

$$\mu_\mathrm{LH}(t) = a_\mathrm{f} - a(\boldsymbol{w}(t)), \qquad \mu_\mathrm{HL}(t) = a(\boldsymbol{w}(t)) + a_\mathrm{f}, \tag{4}$$

which measure the distance of the instantaneous drive from the two folds and constitute the control-relevant output of the identified model.

2.3 Observation model and regime probability

The latent state is linked to a measured emission proxy $y(t)$ (bright in state L, quiescent in state H) through a learned map $\hat{y} = h_\psi(\beta(\zeta), \boldsymbol{w}, u_\mathrm{edge}, n_\mathrm{edge})$. Crucially, y never enters the latent dynamics (3); it only supervises the observation head, which breaks the circularity of proxy-following formulations. The instantaneous regime probability is the calibrated logistic

$$p_\mathrm{H}(t) = \sigma\Big(k\,\zeta(t)/\sqrt{b}\Big)\,, \tag{5}$$

and the mathematically sharp classification is basin membership: the system is in regime H at time t iff $\zeta(t)$ lies in the basin of attraction of $\zeta^*_\mathrm{H}\big(a(t)\big)$.

3 Identification

Data. Publicly released measurement campaigns from a toroidal confinement experiment (MAST) (UK Atomic Energy Authority, 2024); each discharge provides profile measurements $u(\rho_i, t_j)$ in the outer domain, exogenous inputs $\boldsymbol{w}(t)$, and the emission proxy $y(t)$ at millisecond resolution. The corpus contains 36 discharges: 26 with at least one L→H transition (several with back-transitions or dithering phases) and 10 that remain in state L throughout, retained deliberately as negative examples. Discharges are screened by strict per-channel quality gates (median level, bounded jumps and span, minimum time coverage), which is essential: raw channels are frequently corrupted, and admitting them destabilizes both the rollout and the labels.

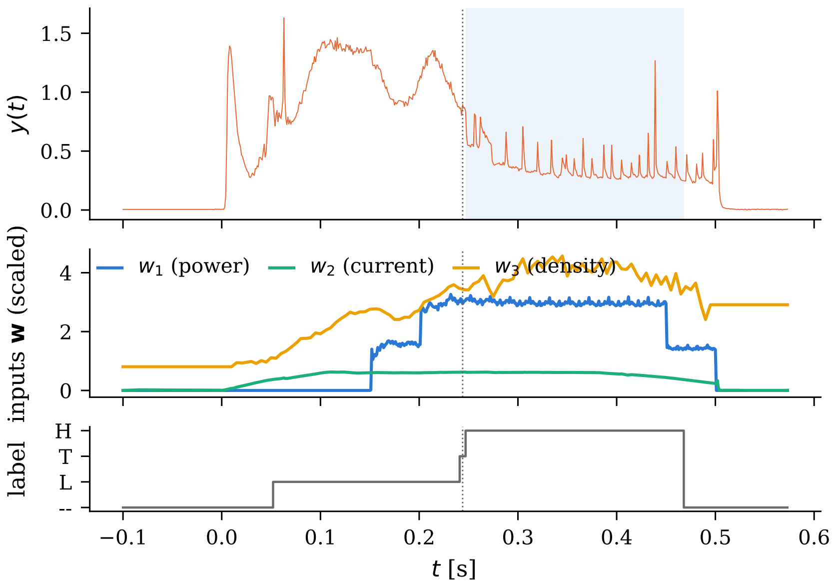

Figure 1: Example discharge. Top: emission proxy $y(t)$; the shaded band is the weakly labeled H segment and the dotted line the labeled transition time. Middle: exogenous inputs in fixed physical scales. Bottom: weak segment labels (L / transition / H; "–" marks the excluded formation and termination phases).

Weak segment labels. Because y is bright in state L and quiescent in state H but heavily contaminated by short bursts, labels are computed from its *lower envelope* (rolling 15th percentile, 15 ms window). Within a gate that excludes the formation and termination phases of each discharge, the envelope distribution is split by a variance-maximizing threshold; a candidate H segment is accepted only if (i) it persists for at least 20 ms, (ii) it is preceded by an L reference inside the gate, and (iii) the envelope steps down sharply across its entry (25 ms contrast windows), which rejects slow drifts. Detected transition times agree with independent session-log annotations to within $\sim$20 ms on the discharges where such annotations exist.

Objective. With $\Theta = (\theta, \psi, \boldsymbol{\kappa}, b, \tau, k, \chi\text{-parameters})$, the per-discharge loss is

$$\mathcal{L} = \underbrace{\sum_{i,j} W_{ij}\,\phi_\delta\big(\hat{u}(\rho_i,t_j)-u_{ij}\big)}_{\text{profile misfit}} + \lambda_s \sum \phi_{\delta_s}(s_\theta) + \lambda_1 \overline{\zeta^2} + \lambda_2 \overline{\dot{\zeta}^2} + \lambda_r\,\mathrm{BCE}\big(p_{\mathrm{H}},\ell\big) + \lambda_y\,\overline{\big(h_\psi-\tilde{y}\big)^2}, \quad (6)$$

where ϕ_δ is the pseudo-Huber function, W inverse-coverage weights that equalize sparsely and densely observed channels, ℓ the weak segment labels (scored only on clean L/H samples), and $\tilde{y}$ the range-normalized emission proxy. We use $\lambda_r = 5 \times 10^{-2}$ and $\lambda_y = 1$; in particular the emission term supervises the *observation head*, not the latent directly. Gradients of (6) are obtained by reverse-mode differentiation through a θ-method implicit–explicit integrator ($\theta = 0.7$, five substeps per observation interval) whose implicit stage is a tridiagonal (Thomas) solve of the diffusion operator; the latent and source terms are explicit. Model selection uses a held-out validation set of discharges whose loss includes the regime and emission terms, so checkpoints are selected for regime fidelity as well as profile fit.

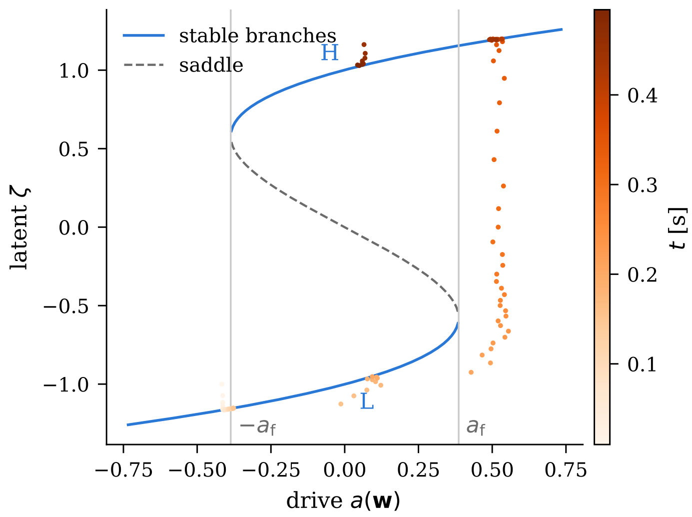

Figure 2: Identified cusp equilibrium manifold (solid: stable branches; dashed: saddle) with the inferred trajectory $\big(a(\boldsymbol{w}(t)), \zeta(t)\big)$ of one discharge overlaid, colored by time. The state tracks the L branch, is carried past the fold a_f by the rising drive, and jumps to the H branch — a saddle-node transition, not a fitted threshold.

4 Results

Identified normal form. Trained on all 36 discharges (20% held out for model selection), the identified latent parameters are $b = 1.00$, $\tau = 23\,\mathrm{ms}$, and fold amplitude $a_f = 0.386$. In physical units the drive map gives a forced-transition threshold ($a = +a_f$) at $\approx 2.7\,\mathrm{MW}$ of injected power at the corpus-typical current and density — an interpretable, testable prediction of the identified model, not an imposed constant. The relaxation time τ matches the observed 20 ms to 40 ms rise of the transition, and the profile misfit in the supervised outer annulus is 73 eV (29% relative).

Quantitative regime assessment. Scoring the calibrated probability $p_H(t)$ against the weak labels on clean L/H samples: over the 26 discharges containing a labeled H segment, the median per-shot AUC is 0.95 (mean 0.85 pooled over all 36), median accuracy 0.87, and the median error between the model's first sustained $p_H = 0.5$ upcrossing and the labeled transition time is 28 ms (mean 45 ms). All 10 non-transitioning discharges — including purely resistively heated ones — are classified L throughout (p_H never crosses 0.5), which is the falsifiability test that proxy-following formulations fail by construction: here the latent is driven only by the actuators, yet it declines to invent transitions where none occur.

5 Discussion

The fold margins of Fig. 4 are the control-relevant output of the identification: $\mu_{LH}(t)$ is the instantaneous distance (in drive units, hence convertible to watts) from a forced L→H transition, and $\mu_{HL}(t)$ the distance from losing the H state. A regulator that holds μ_{HL} positive with a prescribed margin maintains the high-confinement regime with a quantified robustness budget; conversely, scheduling the drive through $+a_f$ at a chosen time forces the transition deliberately. Because both quantities are closed-form functions of the identified parameters and the measured inputs, they are computable in real time.

Limitations. First, the segment labels are weak: they derive from the same emission proxy the observation head predicts (although they never enter the latent dynamics), transition

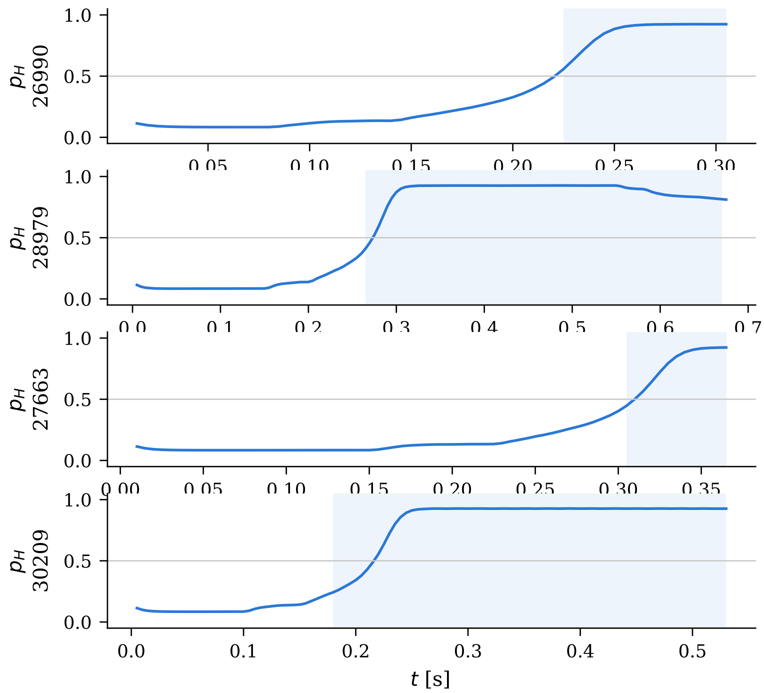

Figure 3: Regime probability $p_{\mathrm{H}}(t)$ (blue) for four discharges; shaded bands are the weakly labeled H segments. The probability rises through the labeled transitions and remains near its branch value between them.

windows are short, and dithering phases are labeled crudely; the classification ceiling is set by label quality. Second, the latent-feature normalization of the legacy designs and the per-shot standardization of the source-network inputs are non-causal (whole-shot statistics); the cusp drive itself uses fixed physical scales and is causal, but a fully real-time implementation must also make the auxiliary inputs causal. Third, a single scalar mode cannot represent oscillatory transition precursors (dithering limit cycles); an extension to a two-dimensional normal form with a Hopf sector is natural future work. Finally, the identified fold threshold is a property of this corpus (one machine, one current/density neighborhood); transfer across operating regimes requires letting $a(\boldsymbol{w})$ be nonlinear in the inputs, for which the affine map here is the first term.

References

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Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press, 2 edition, 2018.

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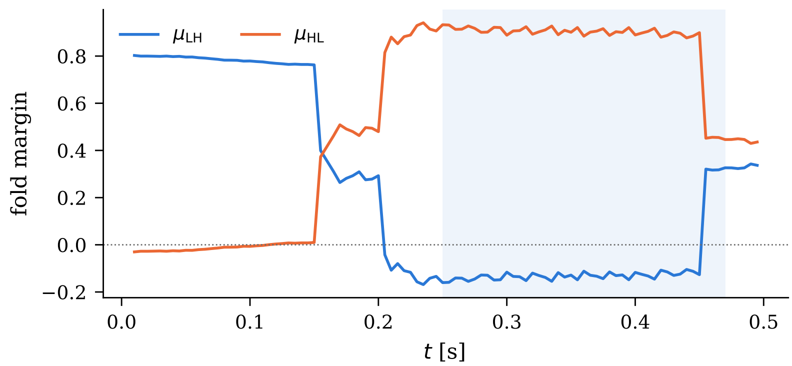

Figure 4: Fold margins along the discharge of Fig. 2. $\mu_{\mathrm{LH}} = a_{\mathrm{f}} - a(t)$ (blue) hits zero at the observed transition — the L branch is annihilated — while $\mu_{\mathrm{HL}} = a(t) + a_{\mathrm{f}}$ (orange) stays positive until the drive is removed, after which the H branch is lost and the state returns to L.